INCH-POUND
A-A-52487
January 24, 1994
SUPERSEDES
MIL-I-62522(AT)
22 July 1987

COMMERCIAL ITEM DESCRIPTION

INSULATION, PANEL NOISE-AND THERMAL-BARRIER

The General Service Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-I-62522(AT), which is cancelled.

ABSTRACT

This CID covers the physical, and performance characteristics of a composite material used to make insulation panels for use in military vehicles. Insulation panels made from this material are used as noise and thermal barriers.

SALIENT CHARACTERISTICS

a. Materials and composition. The insulation composite material shall consist of a noise barrier and a thermal foam barrier. The noise barrier shall completely envelope the thermal foam and, if not, the thermal foam material shall be the closed cell type. The insulation composite material shall not contain any materials which could adversely affect the health of personnel. The use of recovered material made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see note d).

b. Design and construction. The insulation composite material shall be designed and constructed with sound-and thermal-barrier properties for use on metal or plastic panels, and shall satisfy the requirements specified herein.

1. Color and configuration. The color and configuration of the insulation shall be in accordance with the applicable engineering drawing (AED) (see note b).

2. Noise barrier. The noise barrier material shall be self-healing, puncture-resistant, and shall conform to table I.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.

AMSC N/A FSC 5640

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:10Z
Check the source to verify that this is the current version before use.

TABLE I. Noise barrier material.

Characteristic	Requirement
Weight	0.8 pounds per square foot (lb/ft^2) ± 10% loaded plastic having a density of 2.5 pounds per cubic foot (lb/ft^3).
Tensile strength	120 pounds force per square inch (psi) minimum in accordance with ASTM D412, die A.
Tear strength	20 pounds force per inch (lb/in) minimum in accordance with ASTM D412, die C.

3. Thermal foam barrier. The thermal foam barrier material shall be made of urethane foam and shall conform to table II.

TABLE II. Thermal foam barrier.

Characteristic	Requirement
Thermal conductivity	0.30 (Btu)/[(hour) (square ft (ft^2)) (°F/in)] maximum, in conformance with ASTM C177.
Density	2.000 ± 0.2 (lb/ft^3) in accordance with ASTM D297.
Tensile strength	15 pounds per square inch (psi) minimum in accordance with ASTM D3574, die A.
Elongation	200 percent minimum, in accordance with ASTM D412.
Compression set	10 percent maximum at 70 ± 5°F in accordance with ASTM D3574

4. Insulation composite material. The insulation composite material shall conform to table III.

TABLE III. Insulation composite.

Characteristics	Requirements
Adhesion strength barrier to foam	2.2 pounds per linear inch at 180 degree peel in accordance with ASTM D903.
Sound transmission loss	See table IV.
Flammability	3 inch/minute maximum in accordance with SAE J369.
Temperature limits	-40 to 300°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:10Z
Check the source to verify that this is the current version before use.

c. Acoustical performance. The sound transmission loss shall be in accordance with table IV and ASTM E90.

TABLE IV. Acoustical performance.

Frequency (Hz)	Sound transmission loss * (dB)
250	29
500	34
1000	40
2000	45
4000	55

* Of the insulation mounted on 16-gauge sheet steel.

d. Material safety data sheet (MSDS). The contractor shall prepare an MSDS when required to comply with the provisions of FED-STD-313 (see notes b and e).

e. Identification marking. Unless otherwise specified in the AED identification markings shall be permanent and legible. The following information shall be included as a minimum:

1. Manufacturer's CAGE Code.
2. Contract number.
3. Date of manufacturer.
4. AED part number.

QUALITY ASSURANCE PROVISIONS

a. Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

b. Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

PRESERVATION, PACKAGING, PACKING, AND MARKING

Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see b).

NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

a. Addresses for obtaining copies of referenced documents.

1. Government specifications and standards. Copies of FED-STD-313, "Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities" are available from the Navy Publishing and Printing Service Office, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:10Z
Check the source to verify that this is the current version before use.

2. Non-Government publications. ASTM C117 “Steady-State Thermal Transmission Properties by Means of the Guarded Hot Plate”; ASTM D297 “Rubber Products-Chemical Analysis”; ASTM D412 “Rubber Properties in Tension”; ASTM D903 “Peel or Stripping Strength of Adhesive Bonds”; ASTM D3574 “Flexible Cellular Materials-Slab, Bonded, and Molded Urethane Foams”; ASTM E90 “Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions”; are available from the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, PA 19103. SAE J369 “Flammability of Polymeric Interior Materials - Horizontal Test Method, Standard” is available from the Society of Automotive Engineers, Inc. (SAE), 400 Commonwealth Drive, Warrendale, PA 15096.

b. Ordering data. Acquisition documents must specify the following:

1. Title, number, and date of this CID.
2. Title, number, and date of AED.
3. Identify activities requiring copies of completed MSDS and specify where the MSDS will be inspected.
4. Selection of applicable level and packaging requirements.
5. Issue of the DoDISS and the Industry Standards to be cited in the solicitation and the specific issue of individual documents referenced.

c. Cross-reference. Insulation conforming to this CID are interchangeable/substitutable with insulation conforming to MIL-I-62522(AT).

d. Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

e. MSDS. The contracting officer should identify those activities requiring copies of the completed MSDS prepared in accordance with FED-STD-313.

MILITARY INTERESTS:

Custodian:
Army - AT

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Preparing Activity:
Army - AT

(Project 5640-0557)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:10Z
Check the source to verify that this is the current version before use.